

EXECUTIVE SUMMARY

PRAYER PHENOMENA FRAMEWORK (PPF)

*A Phenomenological Paradigm for Determining
Islamic Prayer Times*

Author / Institution	Prayer Phenomena Framework (PPF) Research Project
Date	July 10, 2026
Status	Draft v1.0 — Provisional
Scope	22 Chapters ~10,940 Lines 6 Major Parts

I. BOOK OVERVIEW & THESIS

Prayer Phenomena Framework (PPF):

This book presents a comprehensive re-evaluation of how Islamic prayer times — particularly **Subuh (Fajr/Dawn)** and **Isya (Isha/Twilight)** — are determined. The central thesis challenges the global reliance on **fixed-angle parameters** (e.g., 18° for MWL, 19.5° for EGAS, 20° for Kemenag RI, 15° for ISNA) as representations of dawn/twilight phenomena.

The book argues that these parameters have undergone an "**epistemological rupture**" — the representation (model/angle) has replaced the phenomenon (actual dawn light) as the primary reference, creating a fundamental disconnect between Islamic prayer practice and the natural phenomena that are supposed to govern it.

The book proposes a paradigm shift from **angle-based fixed parameters** to a **brightness-based adaptive methodology** grounded in the Quranic principle of *yatabayyana* (making things clearly visible) and the prophetic criteria of *al-fajr al-mustathir* (the spreading dawn) versus *al-fajr al-mustathil* (the vertical false dawn).

II. STRUCTURAL OVERVIEW — THE SIX PARTS

Part I: Philosophical & Methodological Foundations (Chapters 1–3)

Ch.	Title	Core Contribution
1	Fenomena sebagai Realitas Utama: Merumuskan Paradigma	Establishes that natural phenomena (dawn light) are the primary reality (al-ashl), while mathematical models/angles are mere representations (al-far'). Introduces the concept of "epistemological rupture."
2	Metodologi Evaluasi Representasi: Dari Fenomena ke Model dan Kembali	Builds a three-dimensional evaluation methodology — Physical Accuracy, Phenomenological Fidelity, Shar'i Validity — within a continuous cycle: Observation → Model → Verification → Revision.
3	Integrasi Ushul Fikih, Maqashid Syariah, dan Filsafat Ilmu	Synthesizes Islamic legal epistemology, teleological framework, and philosophy of science into the Three-Dimensional Evaluation Matrix (Shihah-Yusr-'Umum) — MWL 18° scores 3.3/5 and EGAS 19.5° scores 2.3/5.

Part II: Textual Analysis — Quranic & Prophetic Evidence (Chapters 4–6)

Ch.	Title	Core Contribution
4	Fenomena Waktu Shalat dalam Al-Qur'an	Thematic exegesis of 9 Quranic verses. Finds the Quran uses exclusively phenomenological language — no degrees,

		minutes, or quantitative parameters. Three principles: phenomenological, gradual, participatory observer.
5	Fenomena Waktu Shalat dalam Sunnah	Analysis of Hadits Jibril, hadits on al-fajr al-mustathir vs. al-mustathil, isfar vs. taghlis, and syafaq. Confirms hadith describes visual phenomena, not angles.
6	Pemahaman Salaf tentang Subuh dan Isya	Analysis of ~982 early Islamic atsar (reports). Finds zero mention of degree angles in the first three generations. Salaf used purely qualitative, visual descriptions.

Part III: Historical Analysis — From Phenomena to Standardization (Chapters 7–10)

Ch.	Title	Core Contribution
7	Sejarah Konsep Fajar dan Syafak	Traces the conceptual evolution in 6 stages: linguistic → visual → computational → standardized. Documents the critical shift from phenomenological event to quantified parameter.
8	Sejarah Kuantifikasi: Tokoh, Angka, dan Metode	Profiles key figures: al-Farghani, al-Biruni, Ibn al-Shatir, al-Khalili. Shows 18° dominates the classical tradition but was always understood as approximate.
9	Era Standardisasi Modern	Examines institutional standardization: EGAS (19.5°), MWL (18°), ISNA (15°), Umm al-Qura (18.5°). No institution scores >3.5/5, and no documentation exists for empirical basis.
10	Kolonialisme, Ilmu Pengetahuan, dan Falak	Debunks the "colonial conspiracy" narrative. Shows 19.5° is not colonial in origin, but is epistemically unvalidated.

Part IV: Scientific Analysis — Physics, Perception & Computation (Chapters 11–16)

Ch.	Title	Core Contribution
11	Fisika Twilight	Demonstrates that twilight brightness varies 5–10× at the same depression angle depending on atmospheric conditions (aerosols, humidity, altitude). Fixed angles are inherently inadequate.
12	Deteksi Fajar oleh Mata Manusia	Human dawn detection is contrast-based, not absolute-brightness-based. Threshold contrast of 5–15% not yet validated cross-culturally.
13	Variabilitas Geografis dan Lingkungan	Quantifies 12° total variation in equivalent "yatabayyana"

		angle between extreme atmospheric scenarios.
14	Protokol Observasi Fenomena Fajar dan Senja	Develops a standardized observation protocol bridging traditional rukyat with modern astronomy: SQM, spectroscopy, all-sky imaging, blind observation protocols.
15	Pemodelan Komputasional: Dari Sudut ke Kecerahan	Presents the computational paradigm shift. Develops a brightness-based global correction map showing corrections of +3 to +30 minutes vs. the standard 18° model.
16	Validasi Model Terhadap Observasi	Brightness-based model achieves ~80–88% classification accuracy vs. ~70–82% for angle-based models. Dataset remains limited (<150 sessions, 3 locations).

Part V: Evaluative Synthesis — Comparative Evaluation & Case Studies (Chapters 17–19)

Ch.	Title	Core Contribution
17	Evaluasi Komparatif Model Kontemporer	Comprehensive evaluation of 9 contemporary models (MWL, EGAS, ISNA, Umm al-Qura, FCNA, Kemenag, and 3 PPF proposals). All evaluations are provisional.
18	Studi Kasus Penerapan Model	Massive simulation: 9 models × 12 locations × 4 seasons × 3 atmospheric conditions. PPF-CS-0001 ranks #1 in phenomenological fidelity. Critical node — all subsequent chapters depend on it.
19	Implikasi Praktis dari Studi Kasus	Demonstrates that global parameter uniformity is impossible. EGAS 19.5° and Kemenag 20° most in need of reevaluation.

Part VI: Synthesis, Recommendations & Outlook (Chapters 20–22)

Ch.	Title	Core Contribution
20	Sintesis Temuan: Paradigma Fenomenologis untuk Falak Syar'i	Synthesizes all findings into a cohesive Phenomenological Paradigm. Extracts 6+ enduring principles. Verifies 15 hypotheses across 5 disciplines.
21	Rekomendasi untuk Peneliti, Institusi, dan Pemangku Kebijakan	Categorized recommendations: Immediate (0–2 yrs), Short-term (2–5 yrs), Long-term (5–10 yrs). Each tagged by evidence strength.
22	Penutup: Keterbatasan dan Agenda	Documents systematic limitations. Identifies 10+ future

	Riset Masa Depan	research agendas. Commits to the "living research" principle.
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III. KEY FINDINGS

A. Philosophical Finding

Parameters have replaced phenomena. Angle-based parameters (18°, 19.5°, 20°) have been treated as definitions of dawn rather than approximations — a reversal of the proper hierarchy where phenomena are *al-ashl* (primary) and models are *al-far'* (derivative).

B. Textual Finding

No Quranic verse or hadith specifies degree angles. Both the Quran and Sunnah use exclusively phenomenological language — visual descriptions of light, darkness, and contrast. The *yatabayyana* (becoming clearly visible) principle in Al-Baqarah 2:187 and the *al-fajr al-mustathir* (spreading dawn) criterion in the hadith of Ibn Abbas both point to brightness/contrast-based detection, not geometric angle measurement.

C. Scientific Finding

Twilight brightness varies 5–10× at the same depression angle. Atmospheric conditions (AOD, humidity, altitude, light pollution) create massive variations in when dawn actually becomes visible — variations larger than the differences between competing angle parameters (18° vs. 19.5° vs. 20°). Any fixed angle is therefore fundamentally incapable of representing dawn universally.

D. Computational Finding

Brightness-based models outperform angle-based models with ~80–88% vs. ~70–82% classification accuracy, but the dataset remains limited (<150 sessions, 3 locations). Results are "provisionally supported" and require larger-scale validation.

E. Institutional Finding

No current institution scores above 3.5/5 on the three-dimensional evaluation matrix (Shihah-Yusr-'Umum). The EGAS parameter (19.5°) scores lowest at 2.3/5, primarily because it appears to mark zodiacal light rather than true dawn (per Semeida & Hassan, 2018 — a single, unreplicated study).

IV. THE THREE-DIMENSIONAL EVALUATION MATRIX

The book's primary analytical tool is the **Shihah-Yusr-'Umum Matrix**:

Dimension	Source	Core Question	Alignment
Shihah (Validity)	Hifz al-Din	Does the model technically reproduce the Shar'i phenomenon?	Kuhn's Accuracy, Popper's Falsification
Yusr (Ease)	Hifz al-Nafs	Does the model avoid undue hardship?	Kuhn's Simplicity
'Umum (Universality)	'Umum al-Risalah	Does the model work across	Kuhn's Scope

		geographies and demographics?	
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This matrix is applied throughout the book to evaluate all 9 contemporary models.

V. PROPOSED ALTERNATIVE: BRIGHTNESS-BASED ADAPTIVE METHODOLOGY

Rather than replacing one fixed angle with another, the book proposes:

- **Brightness-based thresholds** tied to sky luminance (cd/m^2) and contrast ratios rather than fixed depression angles
- **Atmospheric correction factors** incorporating AOD (Aerosol Optical Depth) data
- **Regional adaptation** acknowledging that the equivalent "yatabayyana" angle varies from $\sim 12^\circ$ (tropical, high-AOD) to $\sim 20^\circ$ (clean atmosphere, moderate latitude)
- **Continuous validation** through standardized observation protocols
- **Institutional flexibility** allowing different regions to adopt locally calibrated parameters while sharing a common methodological framework

VI. CURRENT LIMITATIONS & HONEST CAVEATS

The book is transparent about its limitations:

Limitation	Impact	Mitigation
Validation dataset is small (<150 sessions, 3 locations)	Results are provisional	Flagged explicitly; larger validation is recommended
Semeida & Hassan (2018) finding about EGAS = zodiacal light is unreplicated	EGAS evaluation is fragile	Marked as "based on one study awaiting replication"
Brightness-based approach requires instruments not widely available	Implementation barrier	Acknowledged; long-term research agenda for accessible tools
Cross-cultural validation of contrast thresholds is absent	Generalizability unknown	Identified as future research priority
Author conflict of interest (developed PPF-CS-0001 model)	Potential bias	Explicitly declared; independent evaluation recommended

VII. KEY RECOMMENDATIONS

Immediate (0–2 years)

- Conduct **multi-site observational studies** to validate brightness-based thresholds across diverse geographies
- Establish **documentation requirements** for all institutional prayer time parameters
- Initiate **cross-madhab scholarly dialogue** on phenomenological approach

Short-term (2–5 years)

- Develop **open-source brightness measurement tools** accessible to mosques and communities
- Create **regional calibration databases** linking atmospheric conditions to dawn onset
- Reevaluate **EGAS 19.5°** and **Kemenag 20°** parameters through the proposed methodology

Long-term (5–10 years)

- Build **satellite-based real-time dawn detection** systems
- Establish international **Prayer Time Observatory Network**
- Achieve consensus on a **methodology** (not a single parameter) that is globally adaptable

VIII. SIGNIFICANCE & CONTRIBUTION

This book represents the **most comprehensive academic treatment** of the epistemological, theological, historical, scientific, and computational dimensions of Islamic prayer time determination. Its key contributions are:

- **Philosophical:** Establishes a rigorous epistemological hierarchy (phenomena > models) grounded in both Islamic theology and philosophy of science
- **Textual:** Provides the first systematic thematic analysis of all Quranic and hadith sources on prayer times using modern hermeneutical methods
- **Historical:** Documents the complete trajectory from 7th-century phenomenological descriptions to 20th-century standardized parameters
- **Scientific:** Quantifies atmospheric variability's impact on dawn detection and proposes a physically grounded alternative
- **Methodological:** Creates a reusable three-dimensional evaluation framework applicable to any prayer time model
- **Institutional:** Provides evidence-based, tiered recommendations for researchers, fatwa institutions, and policymakers

The book positions itself as "living research" — designed to evolve as new evidence emerges. All claims are tagged by evidence strength, and the framework explicitly welcomes revision when better data becomes available. The final message is not that any single parameter is "wrong," but that the methodology for determining parameters must be transparent, empirically grounded, and continuously validated against the phenomena that the Quran and Sunnah describe.